Abu Dhabi City Campus Daily/Weekly Planner

Week 8: 19th – 23rd October 2026

Subject: Mathematics **Unit Title: Calculus** **Grade: 12**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Average Rate & Instantaneous Rate of Change	I can calculate and interpret average and instantaneous rates of change using graphs, tables, equations, and real-world contexts.	2.1 Clarify mathematical vocabulary and notation using graphs, tables, worked examples, and connections. 2.3 Support decoding of rate-of-change notation using multiple representations.	Related Videos Limit Definition of the Derivative | Calculus Lesson 8 Derivatives Using The Limit Definition Examples | Calculus
Lesson 2: Defining the Derivative	I can define the derivative as the instantaneous rate of change and use the limit definition to find f'(a).	5.3 Build fluency with graduated scaffolds and worked examples. 8.2 Vary challenge level through structured derivative problems.	
Lesson 3: Review – Limits & Continuity	I can use graphs, tables, and algebraic methods to evaluate limits and determine whether a function is continuous at a point or on an interval.	2.1 Clarify limit and continuity vocabulary using graphs, tables, and worked examples. 2.3 Support decoding of limit notation using multiple representations. 5.3 Build fluency through graduated review and worked examples. 9.3 Develop self-assessment through error analysis and reflection.	
Lesson 4: Formative 2 – Discontinuities, Continuity & Asymptotes	I can demonstrate my understanding of discontinuities, continuity, and asymptotes and justify my answers using mathematical evidence.	5.3 Build fluency through independent assessment and targeted review. 8.2 Vary challenge level through differentiated assessment questions.	
Homework	Practice resources posted on Teams/OneNote; targeted AP Calculus practice.		
Teacher's notes/ Reminders	Microsoft Teams and OneNote; lesson PPTs; review sheets; GDC/Desmos; AP Calculus practice resources.		

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Technology Integration	Microsoft Teams; OneNote; GDC/graphing calculator; Desmos; lesson PowerPoints; digital practice and assessment materials.
Cross-curricular links	Physics/Science: interpret rates of change as motion-related quantities. Real-world contexts: motion, temperature, population and changing quantities.
External Assessment Integration	Direct alignment with AP Calculus AB
Wellbeing/ ATL Skills	Thinking Skills: Connect numerical, graphical and analytical representations; select appropriate methods; analyse errors and justify conclusions. Communication Skills: Use correct limit, continuity and derivative notation and explain reasoning clearly. Self-Management Skills: Review independently, manage assessment time, check work and reflect on misconceptions.
Standards	AP Calculus AB: Topic 2.1 Defining Average and Instantaneous Rates of Change at a Point; Topic 2.2 Defining the Derivative of a Function and Using Derivative Notation. Review/assessment: Topics 1.13 Removing Discontinuities; 1.14 Connecting Infinite Limits and Vertical Asymptotes; 1.15 Connecting Limits at Infinity and Horizontal Asymptotes.
Inquiry/ TOK Question	How does the idea of a limit allow us to describe continuity and instantaneous change, even when we cannot simply substitute a value?
Success Criteria (Measurable)	**Lesson 1: Average Rate & Instantaneous Rate of Change** • **All students must** calculate average rate of change over an interval and identify instantaneous rate of change at a point. • **Most students should** explain the difference between average and instantaneous rate of change and connect instantaneous rate to the slope of a tangent. • **Some students could** explain how average rate of change approaches instantaneous rate of change as the interval becomes smaller. **Lesson 2: Defining the Derivative** • **All students must** identify the derivative as the slope of the tangent line and use correct derivative notation. • **Most students should** use the limit definition to calculate f'(a) for polynomial functions. • **Some students could** derive a general expression for f'(x) using the limit definition and explain its graphical meaning. **Lesson 3: Review – Limits & Continuity** • **All students must** evaluate basic limits and identify whether a function is continuous or discontinuous at a given point. • **Most students should** use the three conditions for continuity to justify whether a function is continuous. • **Some students could** solve unfamiliar problems involving parameters, piecewise functions, and continuity and justify their reasoning. **Lesson 4: Formative Assessment 2** • **All students must** identify holes, jumps, vertical asymptotes and horizontal asymptotes and determine whether a function is continuous. • **Most students should** evaluate limits associated with discontinuities and asymptotes and explain their answers using appropriate notation. • **Some students could** solve unfamiliar problems involving multiple representations and justify conclusions about continuity, discontinuities and asymptotic behaviour.

National Identity Domain	Respect الاحترام Area: Citizenship Implementation: Develop mathematical responsibility by using evidence, precise notation and logical justification when evaluating limits, continuity and asymptotic behaviour.
	Area: Citizenship
	Implementation: Encourage students to check the reasonableness of mathematical conclusions, learn from errors and communicate evidence-based reasoning clearly.